Thermal Guardian:
An Assistive Wearable System for Home-Based Monitoring and Prevention of Bathing Related Health Risks in Older Adults

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Background

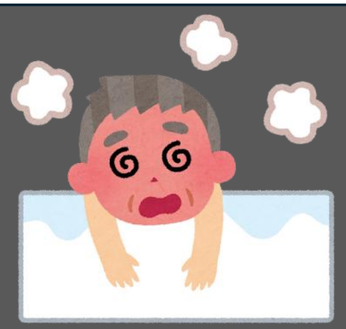
In Japan, hot-bath soaking is a deeply ingrained lifestyle habit, especially in the home

But older adults face bathing-related health risks at home [4]

~19,000 bathing-related deaths occur annually [1]

Issues

Bathrooms are a high-risk environment due to temperature changes [2],[3]



Incidents can happen fast and silently [2], [19]

Constraints:

- Privacy
- Wet/steamy environment
- User variability

Issues

Different phases in bathing = Different physiological changes

Bathroom – dressing room temperature difference → Blood pressure fluctuation

Hot bath immersion → Alters autonomic & hemodynamic regulation

After hot bath immersion → Reduced sympathetic tone

→ Loss of consciousness → Increased risk of silent drowning or collapse

[1],[2],[3],[19]

Related Works

Research appears in the literature as early as 1989 (bathtub ECG feasibility), with continued work through the 2010s–2020s across radar, Wi-Fi, IR, depth sensing, and environmental monitoring.

The three main approaches in the literature are as follows:

- **Kinematic/Activity Monitoring**
- **Physiological Monitoring (ECG/respiration/vital signs)**
- **Environmental Risk Monitoring**

Related Works

Kinematic/Activity Monitoring

Wi-Fi/Radar Monitoring

- IR: Uses IR to detect falls
- Wi-Fi: Uses changes in Wi-Fi signals (CSI) to detect falls
- Radar: Uses low-power radar to measure motion and micro-movements from reflected signals to detect falls

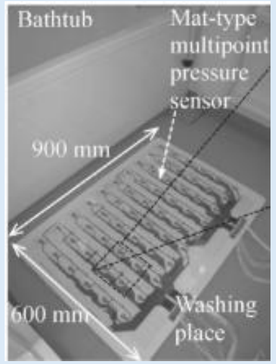
[5],[6],[7],[8]

Camera Detection

- Camera depth: Uses a depth sensor to track body posture/head height and detect possible drowning or collapse

[9],[10],[11]

Pressure Monitoring



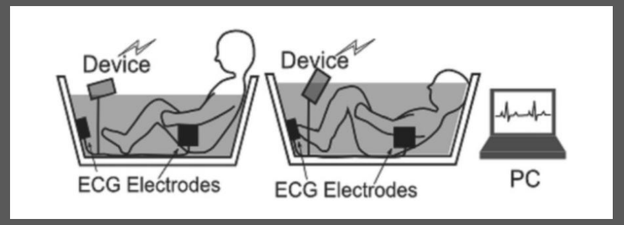
- Pressure mat: Uses a multipoint pressure-sensor bathmat to monitor falls/abnormal events from sudden pressure changes

[12]

Related Works

Physiological Monitoring (ECG/respiration/vital signs)

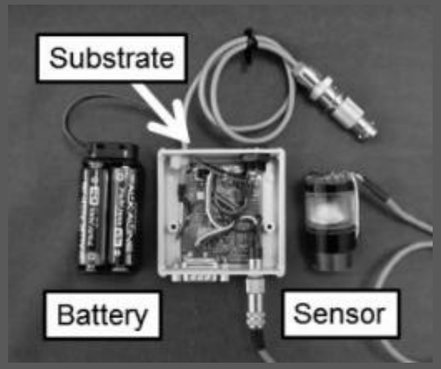
ECG Monitoring



- ECG: Uses electrodes integrated in the bathtub to measure ECG and detect bathing incidents.

[13],[14],[15]

Ultra Sensitive Vibration Monitoring



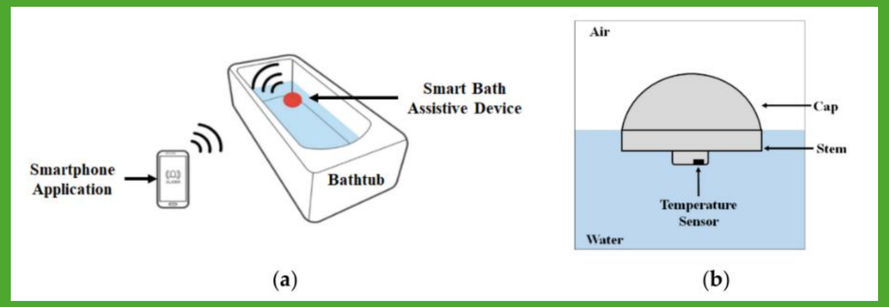
- Places a high-sensitivity vibration sensor on the bathtub to capture tiny vibrations transmitted through the tub, enabling contactless vital-sign (heart-rate) monitoring during bathing.

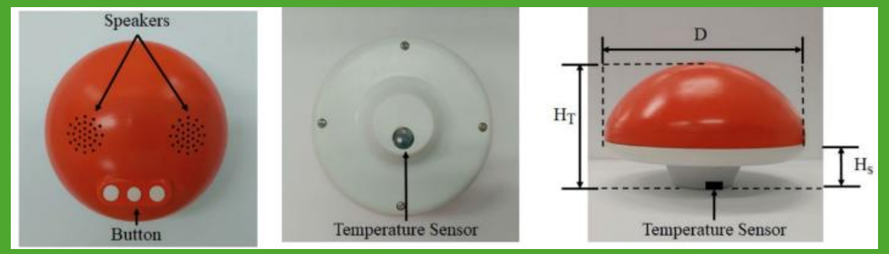
[16]

Related Works

Environmental Risk Monitoring

Water Temperature Monitoring

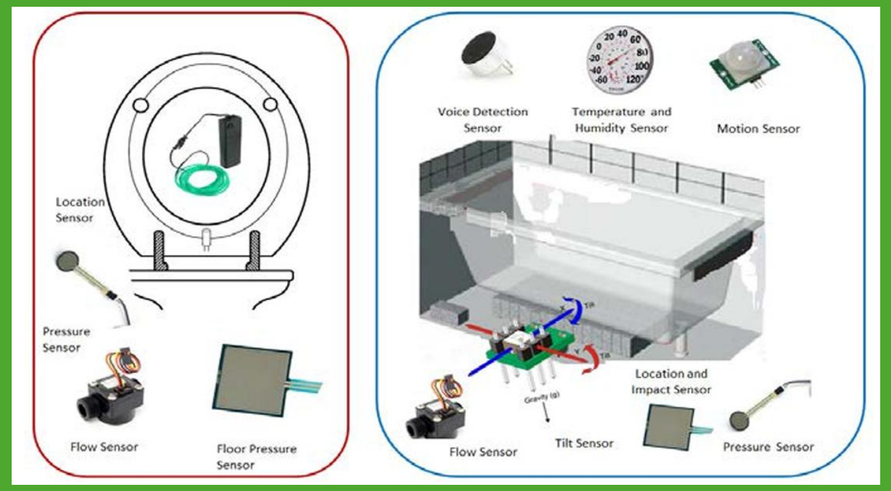




- Measures inner bathwater temperature (not just surface) to track safe bathing conditions and duration

[17]

Ambient Environment Monitoring



- Installs sensors in bathroom fixtures and sends data over a local wireless network to a gateway/cloud for abnormal-event detection and alerts

[16],[18]

Our Approach

- **Combine aspects of all three approaches to make a robust wearable and IoT system that can monitor the user and the environment.**

- **Since bathing accidents can be caused by factors related to the user as well as the environment, monitoring them through a variety of methods will increase chances of prevention/survival of an incident.**

Our Solution

Thermal Guardian

A wearable IoT system designed to enhance bathing safety

Multi-Factor Monitoring
- Physiological
- Posture/Falls
- Environment

Software Support for Caregivers
- Dashboard web app
- Mobile app

Design Considerations

Home-Based

- Designed for daily bathing at home
- Easy and comfortable to wear
- Easy installation

Prevention-Oriented

- Real-time monitoring
- Assistive feedback: haptic & auditory

Context-Aware

- Interpret risk by bathing phase
- User-adaptive algorithm

System Overview

Wearable
HR, SpO2, IMU, Temp, Humidity

IoT Bathroom
Temp, Humidity

IoT Changing Room
Temp, Humidity

Cloud Database
Supabase PostgreSQL

AI Processing
Markov + Safety Rules

Web Dashboard
Facility Staff

Mobile App
Family Caregiver

Push Alerts
Emergency Notifications

Sensors

Wearable:

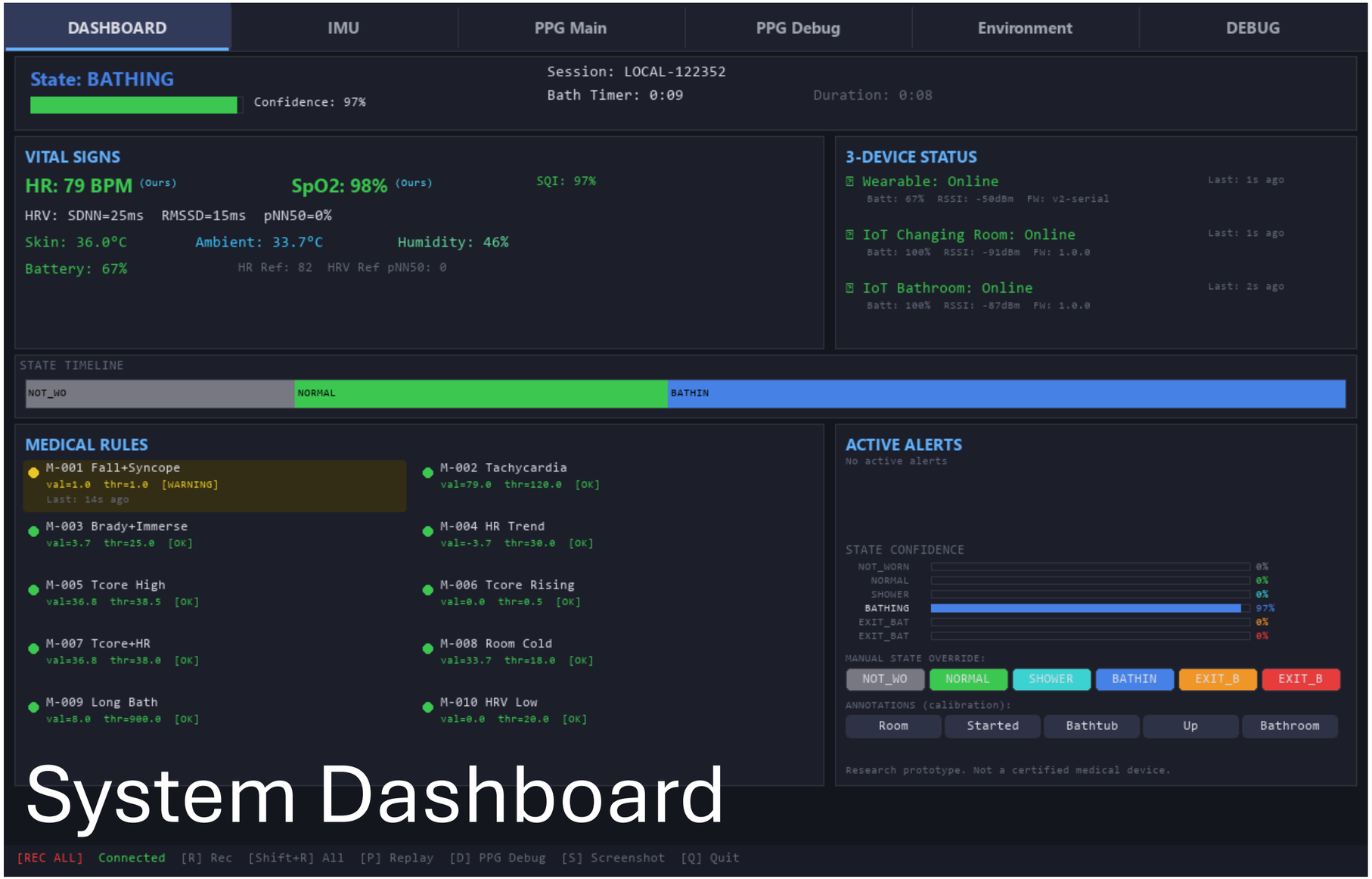
- Heart rate (PPG) sensor
  - Tracks cardiovascular response during bathing
- Movement/activity (IMU sensors)
  - Detects posture / motion change
- Skin temperature sensor
  - Captures thermal load during immersion
- Ambient temperature & humidity sensors
  - Captures user's environment context

IoT Device:

- Ambient temperature & humidity sensors
  - Captures user's environment context

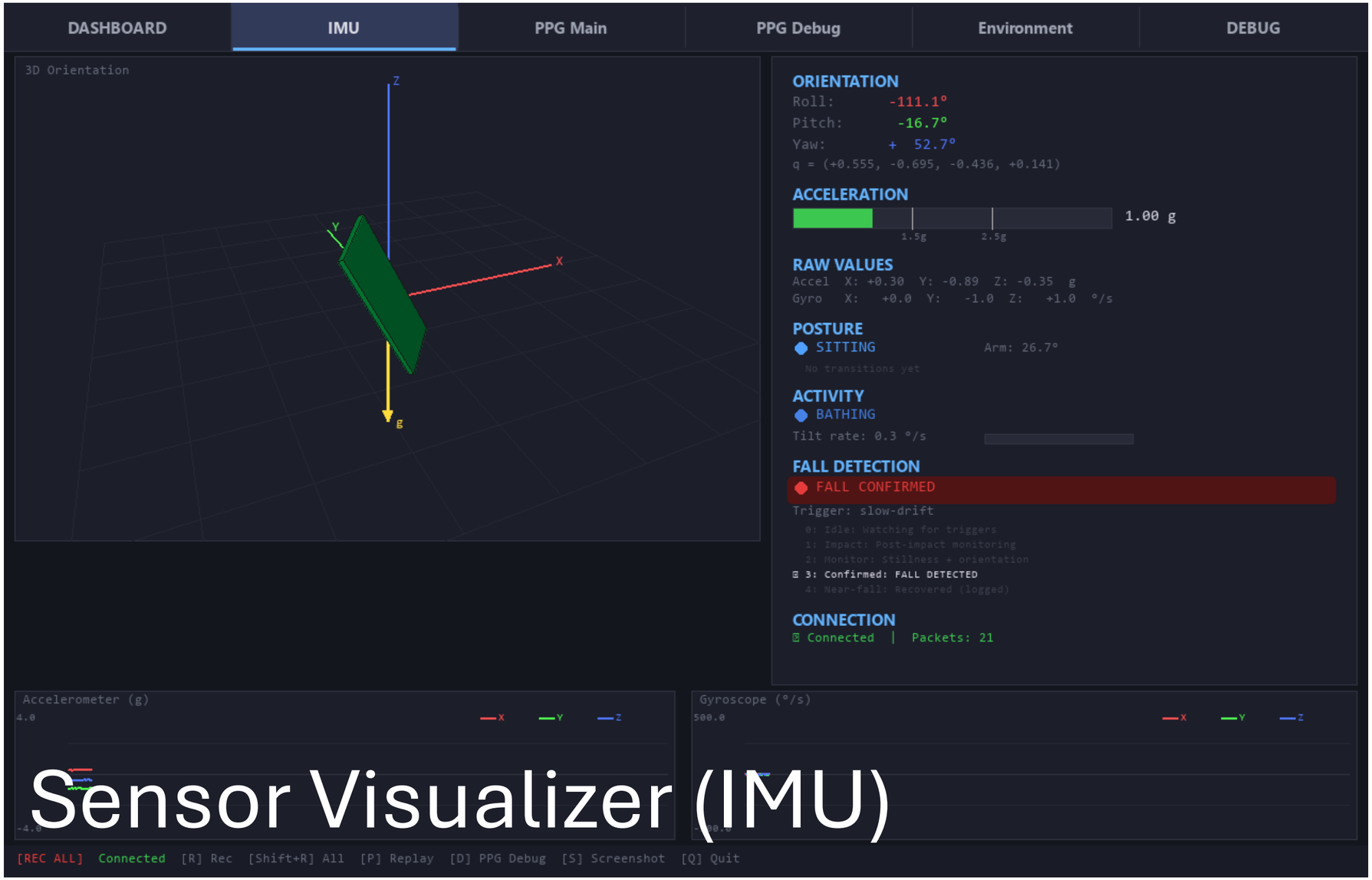
Current Prototype

Current Prototype



System Dashboard

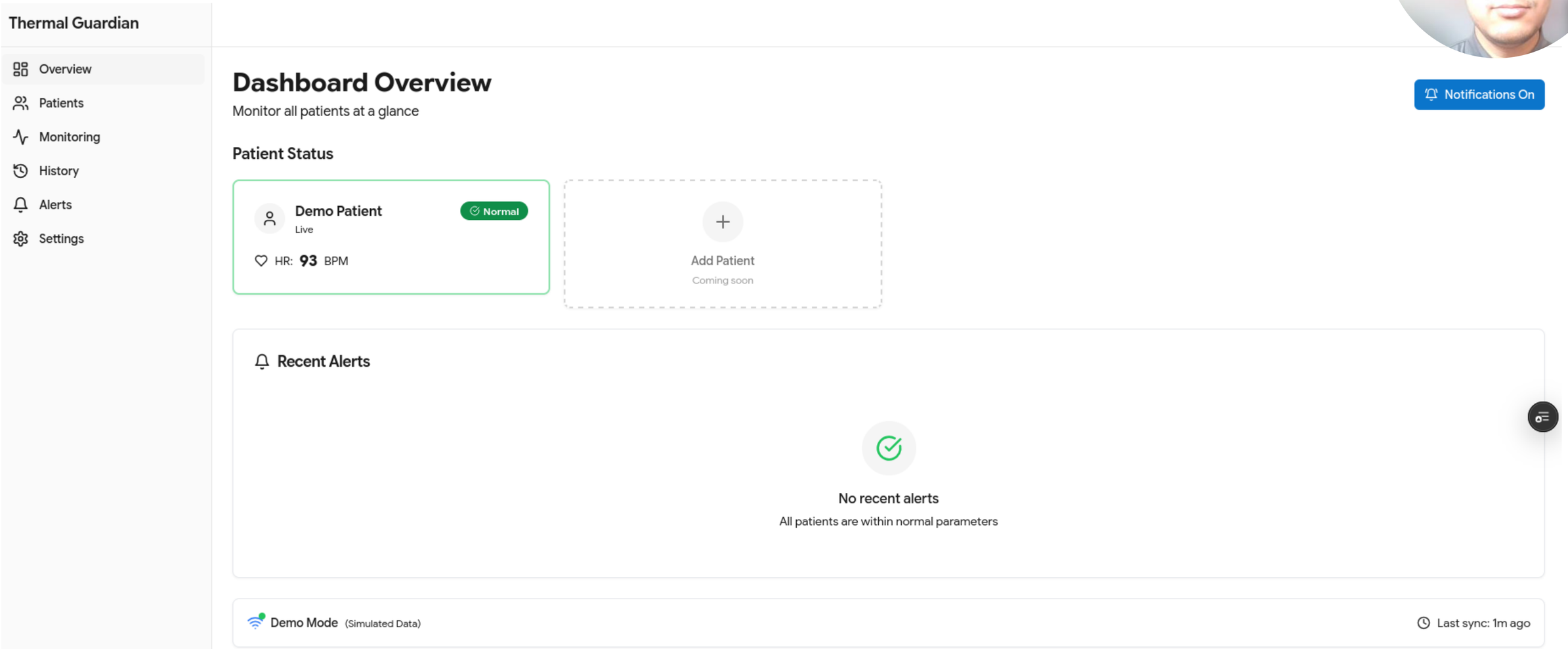
Current Prototype



Sensor Visualizer (IMU)

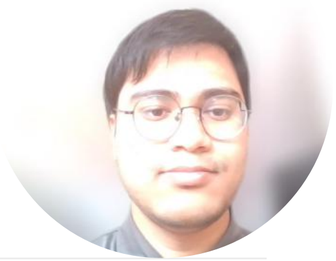
Current Prototype

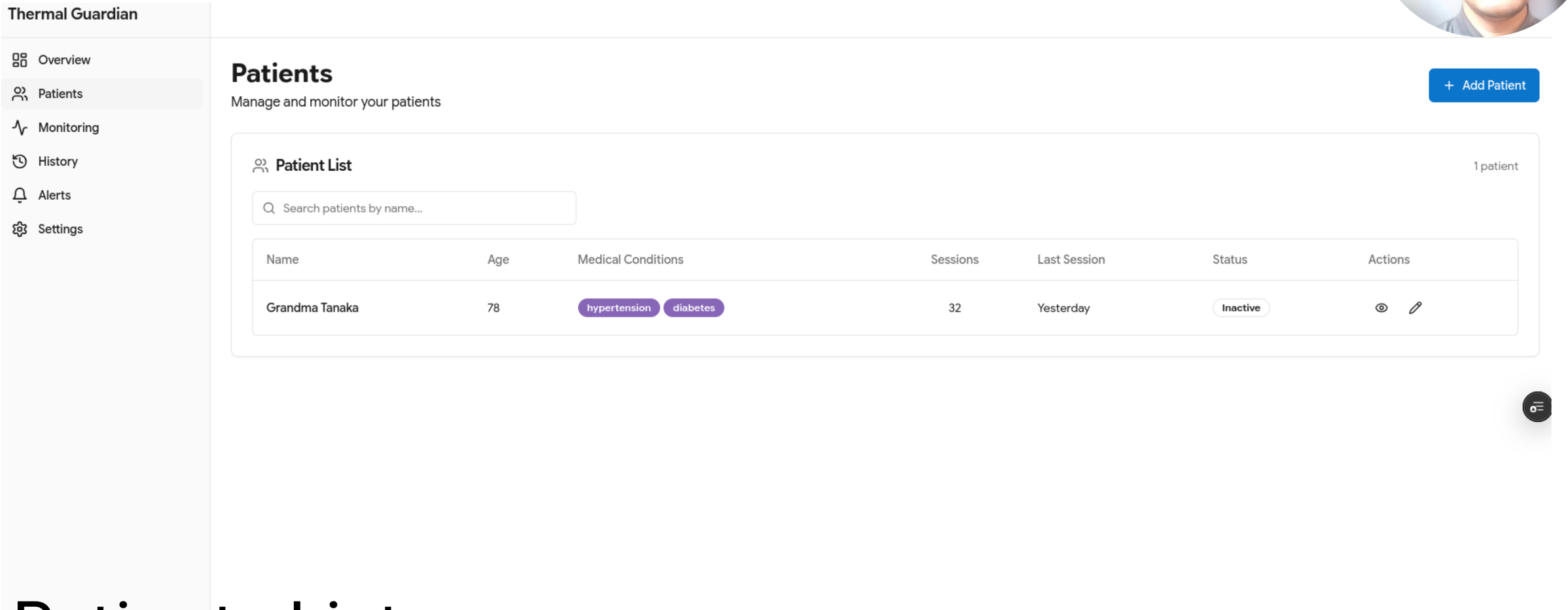
Current Prototype



Web App Dashboard

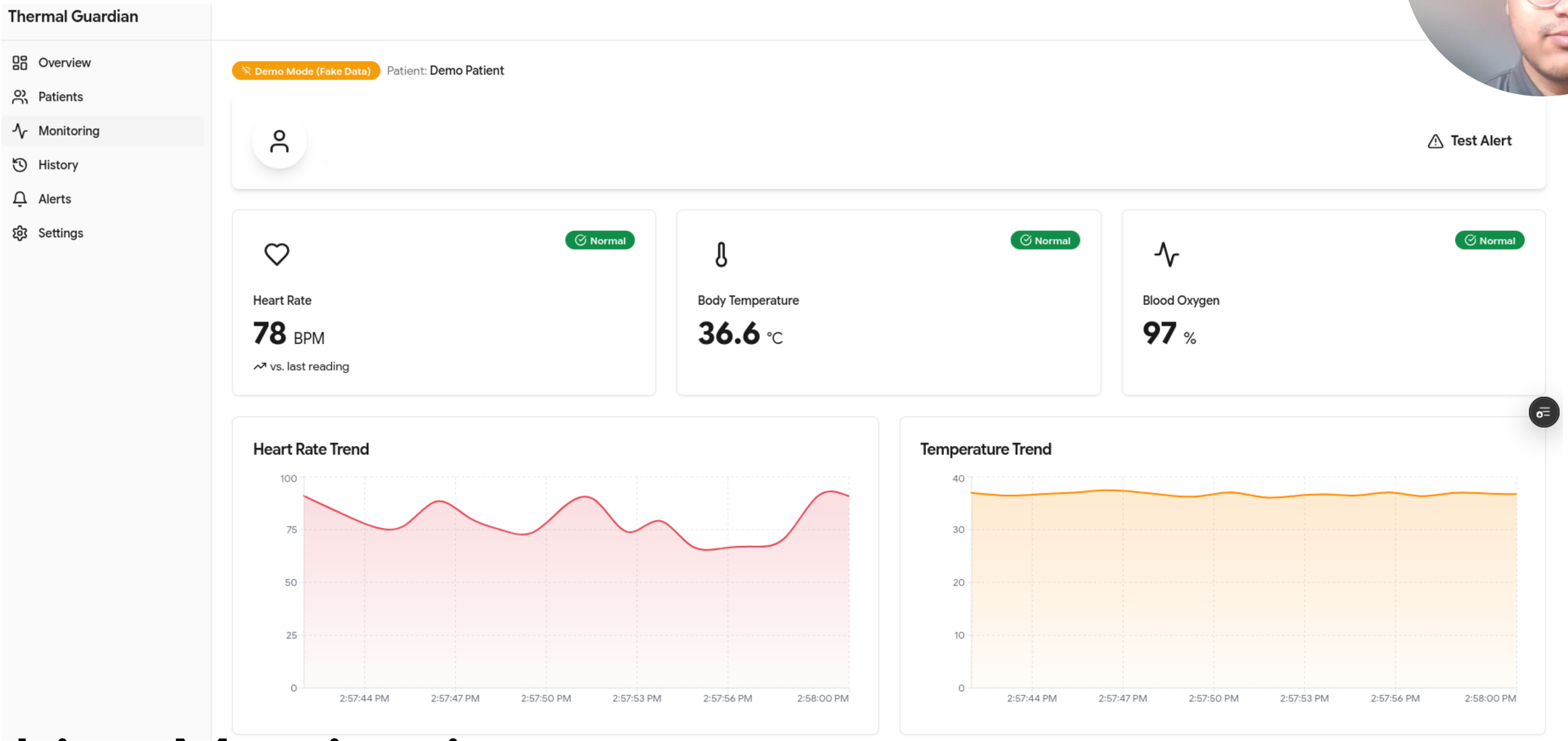
Current Prototype





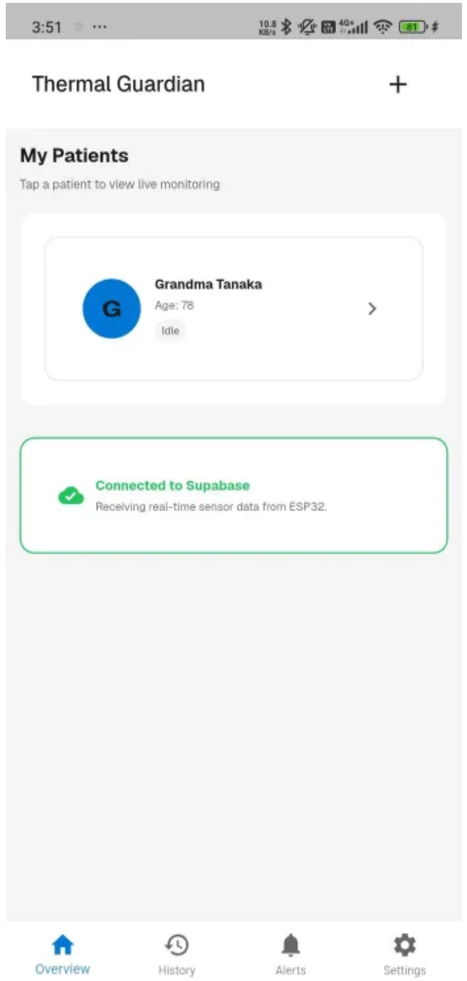
Patients List

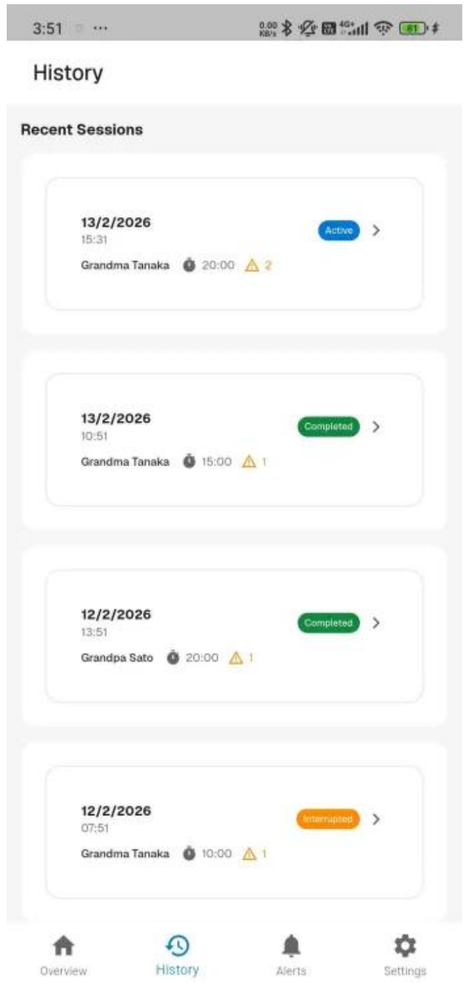
Current Prototype

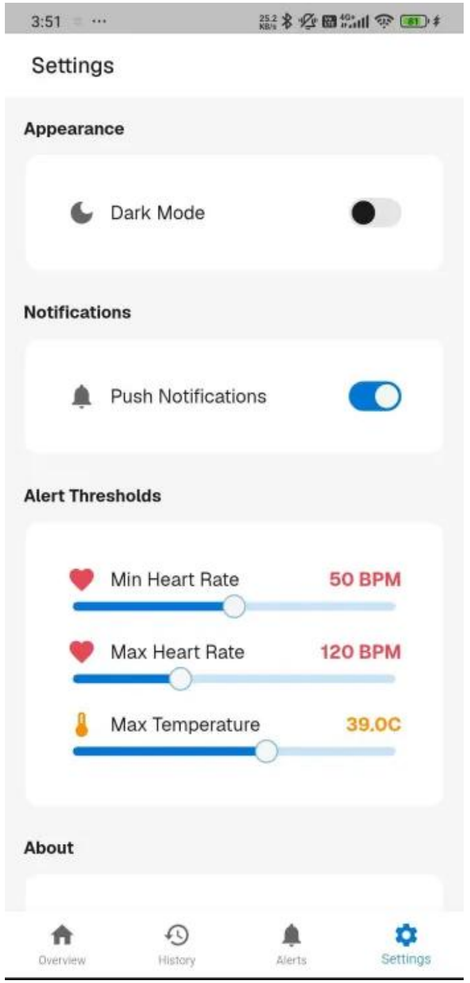


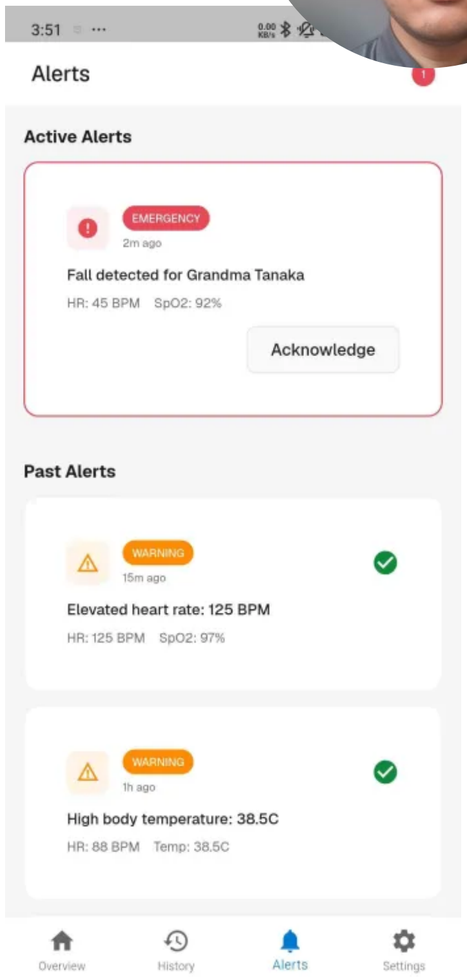
Live Monitoring

Current Prototype









Mobile App

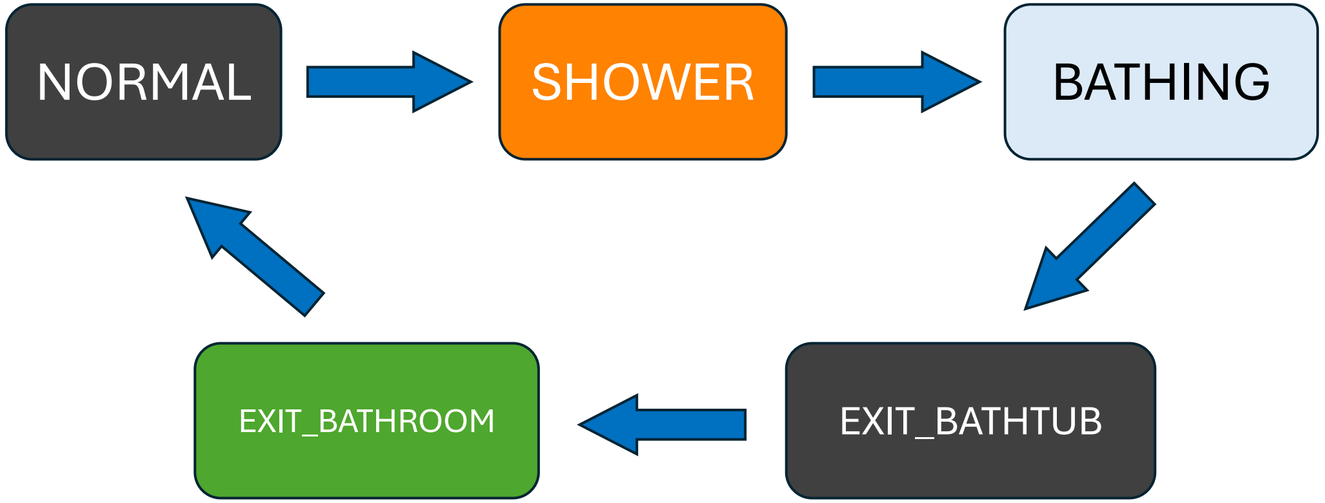
Algorithm: Key Ideas

Same signals ≠ same meaning

- Heart rate/skin temperature changes depend on when they occur in the bathing routine

Use bathing phases as a reference

- Distinguish phases and assess risk accordingly



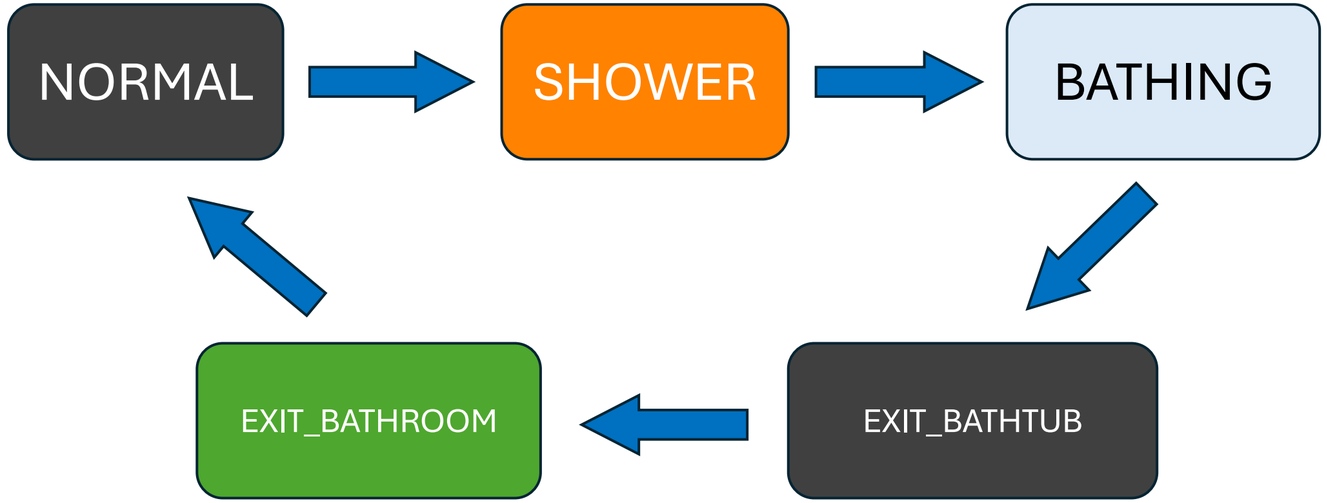
Algorithm: Key Ideas

Different phase; Different risks; Different checks

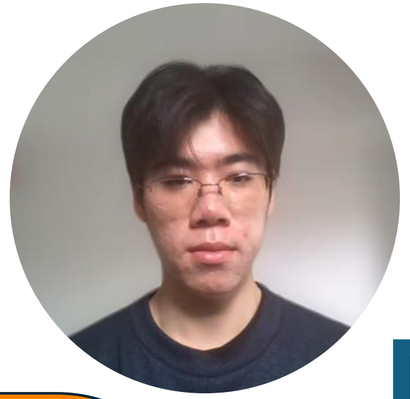
- Only evaluate relevant risks in each phase

More efficient

- Only turn on relevant sensors
- Lower processing load



System Evaluation



Safe Real Bathing

Simulated Bathing

Real bathing test:

- Verify bath-phase detection
- Confirm sensor behavior in real conditions (heat, humidity, immersion)
- Validate safe warning triggers (no dangerous exposure)

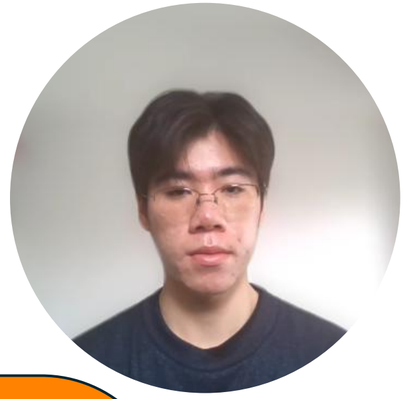
Simulated bathing test:

Validate emergency logic using replay/simulated inputs/test mode

Performance Metrics

Criteria	Method	Threshold
Usability and comfort	User survey	TBD
Algorithm accuracy	Confusion matrix	90% or above on accuracy and precision [20]
Alert delay	Stopwatch	Less than 3 seconds of delay
False alarms	Manual testing	0
Robustness and reliability	Test under various bathing conditions	TBD

Current Limitations



Wearable:

- Waterproofing
- Some failure cases cannot be tested in real subjects due to safety risk, so we rely on simulation/replay
- Comfortability
- Sensor limitations/variability

IoT Device:

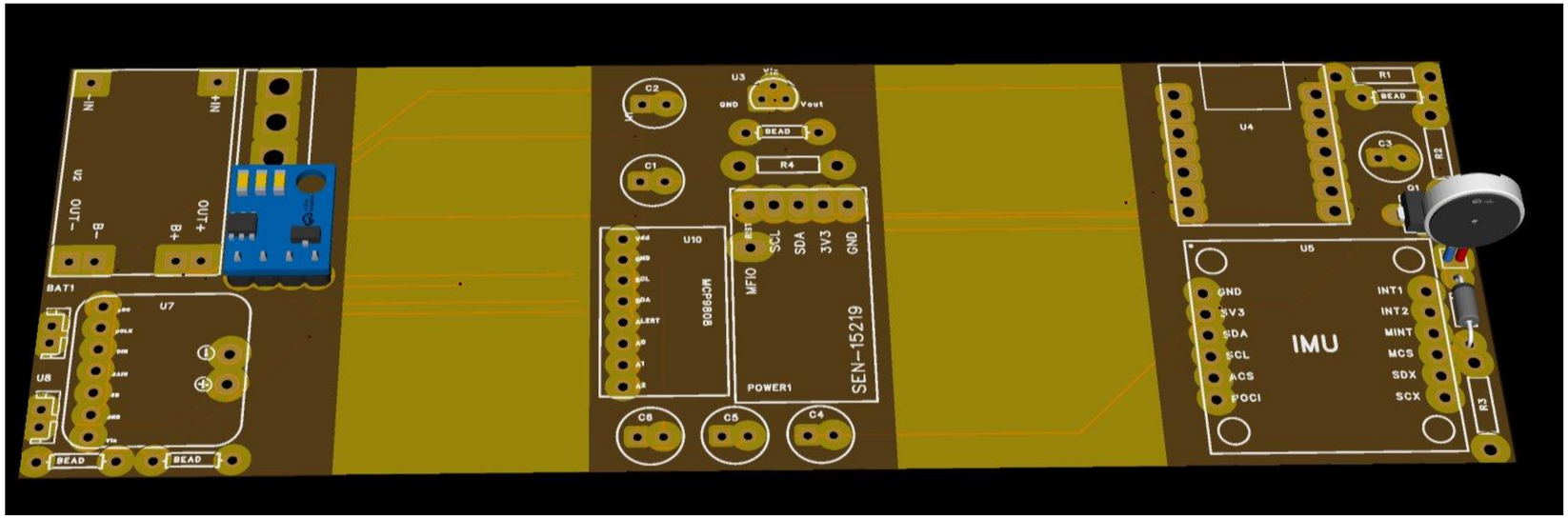
- Not been tested yet

Potential limitations

- Range

Next Steps

- Hardware optimization
  - PCB (Printed Circuit Board)
- Waterproofing & Comfort
  - Silicon casing
- Execute the evaluation plan



Citations

[1] Y. Kawakami, “Deaths in bathtubs in Japan: Forensic and clinical implications,” *J. Nippon Med. Sch. (Nippon Ika Daigaku Zasshi)*, vol. 89, no. 1, pp. 33–39, 2022.
[2] H. Abe, “Reflex syncope during a hot bath as a specific cause of drowning in Japan,” *Journal of Arrhythmia*, vol. 29, pp. 37–38, 2013, doi: 10.1016/j.joa.2012.10.002.
[3] Y. Nagasawa, S. Komori, M. Sato, Y. Tsuboi, K. Umetani, Y. Watanabe, and K. Tamura, “Effects of hot bath immersion on autonomic activity and hemodynamics: Comparison of the elderly patient and the healthy young,” *Japanese Circulation Journal*, vol. 65, pp. 587–592, Jul. 2001.
[4] Y. Tai, K. Obayashi, K. Okumura, Y. Yamagami, and K. Saeki, “Blood pressure, pulse rate, and skin temperature during hot-water bathing in real-world settings among community-dwelling older adults: the HEIJO-KYO Study,” *Environmental Health and Preventive Medicine*, vol. 29, Art. no. 12, 2024, doi: 10.1265/ehpm.23-00320.
[5] P. Duan, J. Li, C. Jiao, Y. Cao, and J. Kong, “WIBFALL: a Device-Free Fall Detection Model for bathroom,” in Lecture notes of the Institute for Computer Sciences, Social Informatics and Telecommunications Engineering, 2022, pp. 182–193. doi: 10.1007/978-3-030-94763-7_14.
[6] Y. Ohnishi, T. Abe, H. Nambo, H. Kimura, and Y. Ogoshi, “Development of Abnormality Detection System for Bathers using Ultrasonic Sensors,” *IEEJ Transactions on Sensors and Micromachines*, vol. 126, no. 12, pp. 662–668, Jan. 2006, doi: 10.1541/ieejsmas.126.662.
[7] K. Jimi, A. Kajiwara, and Y. Yamazaki, “Bathroom monitoring with 79 GHz UWB radar sensor using hidden Markov model,” *IEICE Communications Express*, vol. 9, no. 9, pp. 405–410, Jun. 2020, doi: 10.1587/comex.2020xbl0073.
[8] Z. Zhang, S. Ishida, S. Tagashira, and A. Fukuda, “Danger-Pose detection system using commodity Wi-Fi for bathroom monitoring,” *Sensors*, vol. 19, no. 4, p. 884, Feb. 2019, doi: 10.3390/s19040884.
[9] Y. Endo and C. Premachandra, “Development of a bathing accident monitoring system using a depth sensor,” *IEEE Sensors Letters*, vol. 6, no. 2, pp. 1–4, Dec. 2021, doi: 10.1109/lsens.2021.3139636.
[10] C. He et al., “A Non-Contact fall detection method for bathroom application based on MEMS infrared sensors,” *Micromachines*, vol. 14, no. 1, p. 130, Jan. 2023, doi: 10.3390/mi14010130.
[11] A. Bujnowski, A. Palinski, P. Koscinski, L. Skalski, A. Skurczynska, and J. Wtorek, “Detection of person presence and its activity in the bathtub,” *Journal of Physics Conference Series*, vol. 434, p. 012035, Apr. 2013, doi: 10.1088/1742-6596/434/1/012035.
[12] Motoi, Kosuke & Nakamura, S. & Tanaka, Naoto & Yamakoshi, Ken-ichi. (2013). Development of a new accident detection system in washing place of bathroom using bath mat with multipoint pressure sensor. Transactions of Japanese Society for Medical and Biological Engineering. 51. 268-275. 10.11239/jsmbe.51.268.
[13] M. Ishijima and T. Togawa, “Observation of electrocardiograms through tap water,” *Clinical Physics and Physiological Measurement*, vol. 10, no. 2, pp. 171–175, May 1989, doi: 10.1088/0143-0815/10/2/006.
[14] 優高千穂, 正樹関根, and 俊世田村, “浴槽内心電計と無線型加速度センサを併用した入浴事故検知システムの開発,” 2010 社団法人日本生体医工学会, Dec. 2010, doi: 10.11239/jsmbe.48.584.
[15] K. Motoi, A. Kishimoto, and Y. Yamakoshi, “Development of Nonintrusive Electrocardiogram Monitoring System during Bathing by Only Pasting an Electrode Unit Outside the Bathtub Wall,” *Sensors and Materials*, vol. 37, no. 2, p. 649, Feb. 2025, doi: 10.18494/sam5395.
[16] T. Obo, T. Sawayama, T. Sawayama, and N. Kubota, “Vital Sign Detection System during Bathing using Ultrasensitive Vibration Sensor,” *Transactions of the Institute of Systems Control and Information Engineers*, vol. 30, no. 7, pp. 263–272, Jan. 2017, doi: 10.5687/iscie.30.263.
[17] Q. Wei, S.-M. Kang, and J. H. Lee, “Designing a smart bath assistive device based on measuring inner water temperature for bathing temperature monitoring,” *Sensors*, vol. 20, no. 8, p. 2405, Apr. 2020, doi: 10.3390/s20082405.
[18] D. D. Koo, J. J. Lee, A. Sebastiani, and J. Kim, “An Internet-of-Things (IoT) system development and implementation for bathroom safety enhancement,” *Procedia Engineering*, vol. 145, pp. 396–403, Jan. 2016, doi: 10.1016/j.proeng.2016.04.004.
[19] H. Suzuki, W. Hikiji, T. Tanifuji, N. Abe, and T. Fukunaga, “Characteristics of sudden bath-related death investigated by medical examiners in Tokyo, Japan,” *Journal of Epidemiology*, vol. 25, no. 2, pp. 126–132, 2015.
[20] B. Singh et al., “Real-world accuracy of wearable activity trackers for detecting medical conditions: Systematic review and meta-analysis,” JMIR mHealth and uHealth, https://pmc.ncbi.nlm.nih.gov/articles/PMC11399740/ (accessed Feb. 18, 2026).

Thank You!!